Atiyah Macdonald exercises

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1 Rings and Ideals

Exercise 1.12.

1. $\mathfrak{a} \subseteq (\mathfrak{a} : \mathfrak{b})$
2. $(\mathfrak{a} : \mathfrak{b})\mathfrak{b} \subseteq \mathfrak{a}$
3. $((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c}) = (\mathfrak{a} : \mathfrak{b}\mathfrak{c}) = ((\mathfrak{a} : \mathfrak{c}) : \mathfrak{b})$
4. $(\bigcap_i \mathfrak{a}_i : \mathfrak{b}) = \bigcap_i (\mathfrak{a}_i : \mathfrak{b})$
5. $(\mathfrak{a} : \sum_i \mathfrak{b}_i) = \bigcap_i (\mathfrak{a} : \mathfrak{b}_i)$.

Solution:

1. For $a \in \mathfrak{a}$, $a\mathfrak{b} \subseteq \mathfrak{a}$ since $\mathfrak{a}$ is an ideal.
2. Since $xb \in (\mathfrak{a} : \mathfrak{b})\mathfrak{b}$ iff $x\mathfrak{b} \subseteq \mathfrak{a}$ and $b \in \mathfrak{b}$, $xb \in \mathfrak{a}$.
3. Since $\mathfrak{b}\mathfrak{c} = \mathfrak{c}\mathfrak{b}$ we only need to show the first equality. Observe that $x \in ((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c})$ iff $x\mathfrak{c} \subseteq (\mathfrak{a} : \mathfrak{b})$, that is for all $xc \in x\mathfrak{c}$ with $c \in \mathfrak{c}$, $xc\mathfrak{b} \subseteq \mathfrak{a}$. This is true iff $x\mathfrak{c}\mathfrak{b} \subseteq \mathfrak{a}$.
4. $x\mathfrak{b} \subseteq \bigcap_i \mathfrak{a}_i$ iff $x\mathfrak{b} \subseteq \mathfrak{a}_i$ for all i.
5. Note $x \sum_i \mathfrak{b}_i = \sum_i x\mathfrak{b}_i$. Then $\sum_i x\mathfrak{b}_i \subseteq \mathfrak{a}$ iff $x\mathfrak{b}_i \subseteq \mathfrak{a}$ for all i.

Exercise 1.13.

1. $\sqrt{\mathfrak{a}} \supseteq \mathfrak{a}$
2. $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$
3. $\sqrt{\mathfrak{a}\mathfrak{b}} = \sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}} \cap \sqrt{\mathfrak{b}}$
4. $\sqrt{\mathfrak{a}} = (1) \iff \mathfrak{a} = (1)$
5. $\sqrt{\mathfrak{a} + \mathfrak{b}} = \sqrt{\sqrt{\mathfrak{a}} + \sqrt{\mathfrak{b}}}$
6. if $\mathfrak{p}$ is prime, $\sqrt{\mathfrak{p}^n} = \mathfrak{p}$ for all $n > 0$.

Solution:

1. If $x \in \mathfrak{a}$, then $x^1 \in \mathfrak{a}$ so $x \in \sqrt{\mathfrak{a}}$.

2. $x \in \sqrt{\sqrt{\mathfrak{a}}}$ iff $(x^n)^m \in \mathfrak{a}$ iff $x^{nm} \in \mathfrak{a}$ iff $x \in \sqrt{\mathfrak{a}}$.

3. If $x^n \in \mathfrak{ab}$ then $x^n \in \mathfrak{a} \cap \mathfrak{b}$. If $x^n \in \mathfrak{a} \cap \mathfrak{b}$ then $x^n \in \mathfrak{a}$ and $x^n \in \mathfrak{b}$. If $x^n \in \mathfrak{a}$ and $x^m \in \mathfrak{b}$ then $x^{mn} \in \mathfrak{ab}$.

4. If $1^n \in \mathfrak{a}$ then $1 \in \mathfrak{a}$. The converse comes from 1.

5. By 1., $\sqrt{\mathfrak{a}+\mathfrak{b}} \subseteq \sqrt{\sqrt{\mathfrak{a}}+\sqrt{\mathfrak{b}}}$. If $x^n = y+z$ where $y^k \in \mathfrak{a}$ and $z^m \in \mathfrak{b}$ then $x^{n(k+m)} = (y+z)^{k+m}$ has every term in its binomial expansion with either its y term at least k or its z term at least m, and so $x^{n(k+m)} \in \mathfrak{a}+\mathfrak{b}$.

6. If $x^m \in \mathfrak{p}^n$ then $x^m \in \mathfrak{p}$ so $x \in \mathfrak{p}$. Conversely if $x \in \mathfrak{p}$ then $x^n \in \mathfrak{p}^n$ so $x \in \sqrt{\mathfrak{p}^n}$.

Exercise 1.18. If $\mathfrak{a}_1, \mathfrak{a}_2$ are ideals of A and if $\mathfrak{b}_1, \mathfrak{b}_2$ are ideals of B, then

$$\begin{array}{cc}
(\mathfrak{a}_1+\mathfrak{a}_2)^e = \mathfrak{a}_1^e + \mathfrak{a}_2^e, & (\mathfrak{b}_1+\mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c, \\
(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e, & (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c = \mathfrak{b}_1^c \cap \mathfrak{b}_2^c, \\
(\mathfrak{a}_1\mathfrak{a}_2)^e = \mathfrak{a}_1^e\mathfrak{a}_2^e, & (\mathfrak{b}_1\mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c\mathfrak{b}_2^c, \\
(\mathfrak{a}_1 : \mathfrak{a}_2)^e \subseteq (\mathfrak{a}_1^e : \mathfrak{a}_2^e), & (\mathfrak{b}_1 : \mathfrak{b}_2)^c \subseteq (\mathfrak{b}_1^c : \mathfrak{b}_2^c), \\
\sqrt{\mathfrak{a}}^e \subseteq \sqrt{\mathfrak{a}^e} & (\sqrt{\mathfrak{b}})^c = \sqrt{\mathfrak{b}^c}.
\end{array}$$

Solution: Note that extension is monotone. Hence $\mathfrak{a}_1^e \subseteq (\mathfrak{a}_1+\mathfrak{a}_2)^e$ and $\mathfrak{a}_2^e \subseteq (\mathfrak{a}_1+\mathfrak{a}_2)^e$. Hence $\mathfrak{a}_1^e + \mathfrak{a}_2^e \subseteq (\mathfrak{a}_1+\mathfrak{a}_2)^e$. Conversely, let $\sum_i f(a_i+b_i) \in (\mathfrak{a}_1+\mathfrak{a}_2)^e$. Then $\sum_i f(a_i+b_i) = (\sum_i f(a_i)) + (\sum_i f(b_i)) \in \mathfrak{a}_1^e + \mathfrak{a}_2^e$. Next, notice that $(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e$ and $(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_2^e$. Hence $(\mathfrak{a}_1 \cap \mathfrak{a}2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e$. Next, let $\sum_i f(a_ib_i) \in (\mathfrak{a}_1\mathfrak{a}_2)^e$. Then $\sum_i f(a_ib_i) = \sum_i f(a_i)f(b_i) \in \mathfrak{a}_1^e\mathfrak{a}_2^e$. Conversely let $\sum_i(\sum_j f(a_{ij}))(\sum_k f(b_{ik})) \in \mathfrak{a}_1^e\mathfrak{a}_2^e$. Then $\sum_i(\sum_j f(a_{ij}))(\sum_k f(b_{ik})) = \sum_i \sum_j \sum_k f(a_{ij})f(b_{ik}) = \sum_i \sum_j \sum_k f(a_{ij}b_{ik}) \in (\mathfrak{a}_1\mathfrak{a}_2)^e$. Next, let $\sum_i f(a_i) \in (\mathfrak{a}_1 : \mathfrak{a}_2)^e$ such that $a_i\mathfrak{a}_2 \subseteq \mathfrak{a}_1$. Then $(f(a_i))\mathfrak{a}_2^e = (a_i)^e\mathfrak{a}_2^e = (a_i\mathfrak{a}_2)^e \subseteq a_1^e$ so $f(a_i) \in (\mathfrak{a}_1^e : \mathfrak{a}_2^e)$. Next let $\sum_i f(a_i) \in \sqrt{\mathfrak{b}}^e$ Then for some n_i, $f(a_i)^{n_i} \in \mathfrak{a}^e$ so $\sum_i f(a_i) \in \sqrt{\mathfrak{a}^e}$.

Let $a+b \in \mathfrak{b}_1^c + \mathfrak{b}_2^c$. Then $f(a) \in \mathfrak{b}_1, f(b) \in \mathfrak{b}_2$ so $f(a+b) \in \mathfrak{b}_1 + \mathfrak{b}_2$ hence $a+b \in (\mathfrak{b}_1+\mathfrak{b}_2)^c$. Next let $b \in (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c$. Then $f(b) \in \mathfrak{b}_1 \cap \mathfrak{b}_2$. Hence $b \in \mathfrak{b}_1^c \cap \mathfrak{b}_2^c$. Conversely if $b \in \mathfrak{b}_1^c \cap \mathfrak{b}_2^c$ then $f(b) \in \mathfrak{b}_1$ and $f(b) \in \mathfrak{b}_2$ so $b \in (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c$. Next let $\sum_i a_ib_i \in \mathfrak{b}_1^c\mathfrak{b}_2^c$. Then $f(a_i) \in \mathfrak{b}_1$ and $f(b_i) \in \mathfrak{b}_2$ so $f(\sum_i a_ib_i) \in \mathfrak{b}_1\mathfrak{b}_2$. Hence $\sum_i a_ib_i \in (\mathfrak{b}_1\mathfrak{b}_2)^c$. Next let $b \in (\mathfrak{b}_1 : \mathfrak{b}_2)^c$. Then $f(b)\mathfrak{b}_2 \subseteq \mathfrak{b}_1$, that is for each $c \in \mathfrak{b}_2$, $f(b)c \in \mathfrak{b}_1$. If $c' \in \mathfrak{b}_2^c$ so that $f(c') = c$ then $f(bc') \in \mathfrak{b}_1$, so $bc' \in \mathfrak{b}_1^c$. Hence $b \in (\mathfrak{b}_1^c : \mathfrak{b}_2^c)$. Next, if $b \in (\sqrt{\mathfrak{b}})^c$ so that $f(b)^n \in \mathfrak{b}$, then $f(b^n) \in \mathfrak{b}$ so $b^n \in \mathfrak{b}^c$, so $b \in \sqrt{\mathfrak{b}^c}$. Conversely, if $b \in \sqrt{\mathfrak{b}^c}$ so that $b^n \in \mathfrak{b}^c$, then $f(b)^n \in \mathfrak{b}$ so $f(b) \in \sqrt{\mathfrak{b}}$ so $b \in (\sqrt{\mathfrak{b}})^c$.

Exercise (1). Let x be a nilpotent element of a ring A. Show that $1+x$ is a unit in A. Deduce that the sum of a nilpotent element and a unit is a unit.

Solution: Suppose $x^n = 0$. Notice that

$$(1+x)(\prod_{i=0}^{n}(-1)^i x^i) = 1 + x - x - x^2 + x^2 + x^3 \cdots \pm x^n = 1 \pm x^n = 1.$$

If u is a unit then $u^{-1}x$ is nilpotent, so $1 + u^{-1}x$ is a unit, hence $u + x$ is.

Exercise (2). Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x, with coefficients in A. Let $f = a_0 + a_1x + \cdots + a_nx^n$. Prove that

1. f is a unit $\iff$ a_0 is a unit in A and $a_1, \cdots, a_n$ are nilpotent.
2. f is nilpotent $\iff$ $a_0, a_1, \cdots, a_n$ are nilpotent.
3. f is a zero-divisor $\iff$ there exists $a \neq 0$ in A such that $af = 0$.
4. f is said to be primitive if $(a_0, a_1, \cdots, a_n) = (1)$. Prove that if $f, g \in A[x]$, then fg is primitive $\iff$ f and g are primitive.

Solution:

1. If a_0 is a unit and a_i are nilpotent for $i \geq 1$, then $a_i x^i$ are nilpotent and their sum is nilpotent, hence by (1) f is a unit.

   Conversely suppose f is a unit with inverse g (coefficients indexed with b_i). Then $a_0 b_0 = 1$. Next let $\mathfrak{p}$ be a prime ideal. The map $\varphi : A[x] \to (A/\mathfrak{p})[x]$ has $\varphi(f)\varphi(g) = 1$. Hence the degree of $\varphi(f)$ and $\varphi(g)$ are both 0 since $(A/\mathfrak{p})[x]$ is an integral domain. Thus $\phi(a_i) = 0$ for all $i \geq 1$, that is $a_i \in \mathfrak{p}$. Since this is for any prime ideal, the a_i are in all prime ideals, and hence are nilpotent.

2. Similarly to above, if the a_i are nilpotent, their product with indeterminates are, and their sum is.

   Let f be nilpotent with $f^n = 0$. Then $a_0^n = 0$ so a_0 is nilpotent. By exercise (1), $f + 1$ is a unit, and by part 1 the a_i are nilpotent for $i \geq 1$.

3. Let $g \neq 0$ annihilate f with minimum degree m. Then $b_m a_n = 0$. Hence $a_n g$ has degree less than m, and moreover $a_n g$ annihilates f. Hence $a_n g = 0$. Now suppose for induction that $a_{n-i} g = 0$ holds for all $i \leq r$. Then $a_{n-r-1} b_i = -\sum_{j>i} a_{n-j} b_j = 0$. Hence for all i, $a_i b_n = 0$. Thus $b_n f = 0$.

4. Suppose fg is primitive, and call its coefficients c_i. It is sufficient to show f is primitive. Notice that $(a + b) \subseteq (a) + (b)$. Hence $(c_i) \subseteq \sum_{j \leq i}(a_j b_{i-j}) \subseteq \sum_{j \leq i}(a_j)$ for all i. Thus $(1) = \sum_i (c_i) \subseteq \sum_i (a_j)$, and so f is primitive.

   Conversely suppose f and g are primitive. We will show that $\mathfrak{c} = (c_0, c_1, \cdots, c_k)$ is not contained in any prime ideal of A. Suppose it were contained $\mathfrak{p}$. Then under the map $\varphi : A[x] \to (A/\mathfrak{p})[x]$, $\varphi(c_i) = 0$ and so $\varphi(fg) = 0$. Since $(A/\mathfrak{p})[x]$ is a domain, either $\varphi(f) = 0$ or $\varphi(g) = 0$. However, since $(a_0, a_1, \cdots, a_n) \not\supseteq \mathfrak{p}$ (since $\mathfrak{p}$ is proper), some a_i is not sent to zero. Similarly, some b_i is not sent to zero. This is a contradiction, and so $\mathfrak{c}$ is not contained in any prime ideal. Since every proper ideal is contained in a maximal (and hence prime) ideal, $\mathfrak{c}$ is not proper.

Exercise (3). Generalise the results in Exercise 1 to a polynomial ring $A[x_1, \cdots, x_n]$ in several indeterminates.

Solution: The proofs of part 1, 2, and 4 do not depend on being a single variable, so their proofs can be adapted exactly. We will only show part 3. Let f be a zero-divisor with annihilator g. Let d be greater than the maximum exponent of indeterminates in f. Consider the map $\varphi : A[x_1, \cdots, x_n] \to A[y]$ sending x_i to y^{d^i}. Notice that φ restricted to the set of polynomials with maximum exponent indeterminates less than d is injective. To see this, write the exponent of the image of each monomial in base d, each monomial is mapped distinctly. Then $\varphi(fg) = 0$ so $\varphi(f)$

is a zero-divisor. Hence there is an $a \neq 0$ such that $\varphi(af) = 0$. By injectivity of the restriction, $af = 0$.

Exercise (4). In the ring $A[x]$, the Jacobson radical is equal to the nilradical.

Solution: Since the set of maximal ideals is a subset of the set of prime ideals, the nilradical is contained in the Jacobson radical. Conversely let f be in the Jacobson radical. Then $xf - 1$ is a unit, so by Exercise 1 all coefficients of f are nilpotent. Hence f is nilpotent.

Exercise (5). Let A be a ring and let $A[[x]]$ be the ring of formal power series $f = \sum_{n=0}^{\infty} a_n x^n$ with coefficients in A. Show that

1. f is a unit in $A[[x]] \iff a_0$ is a unit in A.
2. If f is nilpotent, then a_n is nilpotent for all $n \geq 0$. Is the converse true?
3. f belongs to the Jacobson radical of $A[[x]] \iff a_0$ belongs to the Jacobson radical of A.
4. The contraction of a maximal ideal $\mathfrak{m}$ of $A[[x]]$ is a maximal ideal of A, and $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x.
5. Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Solution:

1. If f is a unit with inverse $g = \sum_{n=0}^{\infty} b_n x^n$ then $1 = a_0 b_0$ so a_0 is a unit.

   Conversely, suppose a_0 is a unit. We define the inverse g of f by induction. Let b_0 be the inverse of a_0. Next, assume we have the first n terms of g constructed so that the first n terms in their product is zero. Then $c_n + 1 = \sum_{i=0}^{n+1} a_i b_{n+1-i}$. We want $c_n + 1 = 0$ so we get that $b_{n+1} a_0 = -\sum_{i=1}^{n+1} a_i b_{n+1-i}$. Since a_0 is a unit, we can invert it to get $-b_0 \sum_{i=1}^{n+1} a_i b_{n+1-i}$. This completes the construction of the inverse of f.

2. Let f be such that $f^n = 0$. Then $a_0^n = 0$ and so a_0 is nilpotent. Now suppose for induction that a_i are nilpotent for all $i \leq m$ with exponents n_i. Since the nilradical is an ideal, we can define $g = f - \sum_{i \leq m} a_i x^i$ and g will be nilpotent, say $g^k = 0$. Then $a_{m+1}^k = 0$, and so a_{m+1} is nilpotent.

   The converse does not hold, as we could have a power series where each coefficient is nilpotent, but the exponent for it to go to zero is unbounded. TODO make an actual example.

3. f is in the Jacobson radical of $A[[x]]$ iff $1 - fg$ is a unit for all $g \in A[[X]]$. This is true iff $1 - a_0 b_0$ is a unit for all $b_0 \in A$ by part 1. And finally this is true iff a_0 is in the Jacobson radical of A.

4. Since x is in the Jacobson radical, it is it $\mathfrak{m}$. Then since $A[[x]]/(x) = A$, we see that $\mathfrak{m}/(x) = \mathfrak{m}^c$. Hence $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x. Moreover $A[[x]]/\mathfrak{m} = (A[[x]]/(x))/(\mathfrak{m}/(x)) = A/\mathfrak{m}^c$ and so $\mathfrak{m}^c$ is maximal.

5. Let $\mathfrak{p}$ be a prime ideal of A. Then $\mathfrak{p}^e + (x)$ is an ideal of $A[[x]]$ and moreover $A[[x]]/(\mathfrak{p}^e + (x)) = A/\mathfrak{p}$ so $\mathfrak{p}^e + (x)$ is prime. Moreover $\mathfrak{p}$ is the contraction of $\mathfrak{p}^e + (x)$.

Exercise (6). A ring A is such that every ideal not contained in the nilradical contains a nonzero idempotent (that is, an element such that $e^2 = e \neq 0$). Prove that the nilradical and the Jacobson radical of A are equal.

Solution: The nilradical is contained in the Jacobson radical as every maximal ideal is prime, so we just need to show that the opposite inclusion holds. Suppose a is in the Jacobson radical but not the nilradical. Then (a) is an ideal not contained in the nilradical, and so has a nonzero idempotent $e = ab$ (for some $b \in A$). Note also that since a is in the Jacobson radical, $1 - e$ is a unit with inverse c, so $c - ce = 1$. Then $1 - ce = c$, so $1 = (1 - e)(1 - ce) = 1 - e - ce + ce^2 = 1 - e$. This contradicts e being nonzero! So the Jacobson radical and nilradical are equal.

Exercise (7). Let A be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ (depending on x). Show that every prime ideal is maximal.

Solution: Let $\mathfrak{p}$ be prime. Let $x + \mathfrak{p}$ be a nonzero element. Then $x(x^{n-1} - 1) + \mathfrak{p} = 0 + \mathfrak{p}$, so $x^{n-1} \equiv 1 \pmod{\mathfrak{p}}$. Hence $x + \mathfrak{p}$ is a unit, so $\mathfrak{p}$ is maximal.

Exercise (8). Let A be a ring $\neq 0$. Show that the set of prime ideals of A has minimal elements with respect to inclusion.

Solution: We use Zorn's lemma. Firstly, we know that the set of prime ideals is non-empty since it has a maximal ideal. Let C be a chain in the poset of prime ideals. The intersection $\mathfrak{p} = \cap C$ of this chain is an ideal. Suppose $a, b \notin \mathfrak{p}$. Then there is some $\mathfrak{p}_c \in C$ such that $a, b \notin \mathfrak{p}_c$. Hence, $ab \notin \mathfrak{p}_c$, so $ab \notin \mathfrak{p}$. Thus $\mathfrak{p}$ is prime. Every chain has a lower bound, so Zorn's lemma tells us that there is a minimal element.

Exercise (9). Let $\mathfrak{a}$ be an ideal $\neq (1)$ in a ring A. Show that $\mathfrak{a} = \sqrt{\mathfrak{a}} \iff \mathfrak{a}$ is an intersection of prime ideals.

Solution: If $\mathfrak{a} = \sqrt{\mathfrak{a}}$ then $\mathfrak{a}$ is the intersection of all ideals containing it. Conversely, suppose $\mathfrak{a}$ is an intersection of prime ideals $\mathfrak{p}_i$. Let $x \in \sqrt{\mathfrak{a}}$ so that $x^n \in \mathfrak{a}$. Then for each $\mathfrak{p}_i$, $x^n \in \mathfrak{p}_i$ so $x \in \mathfrak{p}_i$. Hence $x \in \mathfrak{a}$.

Exercise (10). Let A be a ring, $\mathfrak{R}$ its nilradical. Show that the following are equivalent:

1. A has exactly one prime ideal;
2. every element of A is either a unit or nilpotent;
3. $A/\mathfrak{R}$ is a field.

Solution: If A has exactly one prime ideal, it is maximal, since A has to have at least one maximal ideal. Moreover, the nilradical is the intersection of all prime ideals, and hence is this ideal. Since every non unit in A is contained in a maximal ideal, which is the nil radical, every non unit is nilpotent.

Now suppose every element is either a unit or nilpotent. Recall that $A/\mathfrak{R}$ has no nilpotent elements. Let $x \in A/\mathfrak{R}$ be nonzero. Then $x \notin A/\mathfrak{R}$, and hence is a unit in A. Thus $A/\mathfrak{R}$ is a field.

Finally suppose $A/\mathfrak{R}$ is a field. Then $\mathfrak{R}$ is maximal. However, it is equal to the intersection of all prime ideals. Thus it is the only prime ideal in A.

Exercise (11). A ring A is *Boolean* if $x^2 = x$ for all $x \in A$. In a Boolean ring A, show that

1. $2x = 0$ for all $x \in A$;
2. every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field of two elements;
3. every finitely generated ideal is principal.

Solution:

1. If $x^2 = x$ then $x + 1 = (x + 1)^2 = x^2 + 2x + 1 = x + 2x + 1$. Hence $2x = 0$.
2. Every prime is maximal from Exercise (7). Let $\mathfrak{p}$ be such a prime. Let $x + \mathfrak{p}$ be nonzero. Then $x^2 - x \in \mathfrak{p}$. Hence $x - 1 \in \mathfrak{p}$. Thus $x \equiv 1 \pmod{\mathfrak{p}}$, so $A/\mathfrak{p} \cong \mathbb{F}_2$.
3. Let $\mathfrak{a}$ be generated by a_1, a_2. Let $a = a_1 + a_2 - a_1a_2$. Then $a_1a = a_1 + a_1a_2 - a_1a_2 = a_1$ and similarly $a_2a = a_2$. Every $x \in \mathfrak{a}$ can be written as $x = x_1a_1 = x_2a_2 = x_1a_1a + x_2a_2a$. Thus $\mathfrak{a}$ is generated by a.

   If $\mathfrak{a}$ is generated by $a_1, \cdots, a_{n-1}, a_n$, by induction the $a_1, \cdots, a_{n-1}$ generate a principal ideal, and so we can repeat the above argument.

Exercise (12). A local ring contains no idempotents $\neq 0, 1$.

Solution: Let $\mathfrak{m}$ be the maximal ideal. Let e be an idempotent in A. If $e \in \mathfrak{m}$ then $1 - e$ is a unit, as $\mathfrak{m}$ is the Jacobson radical. If c is the inverse, then $1 = c - ce$, and then $c = 1 - ce$. Then $1 = (1 - e)(1 - ce) = 1 - e - ce + ce = 1 - e$, so $e = 0$. If $e \notin \mathfrak{m}$, then e is a unit. Since $e^2 - e = e(e - 1) = 0$, we get that $e = 1$.

Exercise (13). Let K be a field and let Σ be the set of irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. Show that $\mathfrak{a} \neq (1)$.

Then they explain how this leads to a construction of the algebraic closure of a field! You should read it!

Solution: Since K has no nilpotent elements, Exercise (3) tells us that a polynomial in A is a unit if and only if it is a constant unit. However, every $f(x_f)$ is non-constant, and the sum and product of non-constant polynomials over a field will be non-constant. Hence there are no units in $\mathfrak{a}$.

Exercise (14). In a ring A, let Σ be the set of all ideals in which every element is a zero-divisor. Show that the set Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence, the set of zero-divisor in A is a union of prime ideals.

Solution: We apply Zorn's lemma. Firstly, the zero ideal is in Σ so it is non-empty. Let C be a chain. Then the union is an ideal. Every element in the union a zero-divisor, so the union is an upper bound of C. Hence maximal elements exist. Let $\mathfrak{p}$ be a maximal element. Assume $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$. Then $(a) + \mathfrak{p}$ and $(b) + \mathfrak{p}$ are strictly larger then $\mathfrak{p}$ and hence contain non-zero divisors $aa' + p, bb' + p$. However $(aa' + p)(bb' + p) = aba'b' + aa'p + bb'p + p^2$ and this is a zero-divisor. This is a contradiction (if $abc = 0$ then a has annihilator bc)! So $\mathfrak{p}$ is prime.

Exercise (15). Let A be a ring and let X be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of prime ideals of A which contain E. Prove that

1. if $\mathfrak{a}$ is the ideal generated by E, then $V(E) = V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$.
2. $V(0) = X$, $V(1) = \varnothing$.
3. if $(E_i)_{i\in I}$ is any family of subsets of A, then
$$V(\bigcup_{i\in I} E_i) = \bigcap_{i\in I} V(E_i)$$
4. $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$ for any ideals $\mathfrak{a}, \mathfrak{b}$ of A.

These results show that the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the *Zariski topology*. The topological space X is called the *prime spectrum* of A, and is written $\mathrm{Spec}(A)$.

Solution:

1. If $E \subseteq \mathfrak{p}$ then $\mathfrak{a} \subseteq \mathfrak{p}$ since $\mathfrak{a}$ is the smallest ideal containing E. Conversely if $\mathfrak{a} \subseteq \mathfrak{p}$ then $E \subseteq \mathfrak{a} \subseteq \mathfrak{p}$. Hence $V(E) = V(\mathfrak{a})$. Next, if $\mathfrak{a} \subseteq \mathfrak{p}$ then $\sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{p}} = \mathfrak{p}$. Conversely if $\sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$ then $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$. Hence $V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$.
2. Every prime ideal contains 0 and no prime ideals contain 1 as they are proper. Hence $V(0) = X$ and $V(1) = \varnothing$.
3. $\bigcup_{i\in I} E_i \subseteq \mathfrak{p}$ iff $E_i \subseteq \mathfrak{p}$ for each $i \in I$. Hence $V(\bigcup_{i\in I} E_i) = \bigcap_{i\in I} V(E_i)$.
4. If $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$ then $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$. Conversely, if $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ then either $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$, and hence $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$. Hence $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$. Next, if $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ then (as stated) either $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$. Conversely if $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$, then $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$. Hence $V(\mathfrak{a}\mathfrak{b}) \subseteq V(\mathfrak{a}) \cup V(\mathfrak{b})$ and $V(\mathfrak{a}) \cup V(\mathfrak{b}) \subseteq V(\mathfrak{a} \cap \mathfrak{b})$.

Exercise (16). Draw pictures of $\mathrm{Spec}(\mathbb{Z}), \mathrm{Spec}(\mathbb{R}), \mathrm{Spec}(\mathbb{C}[x]), \mathrm{Spec}(\mathbb{R}[x]), \mathrm{Spec}(\mathbb{Z}[x])$.

Solution: I don't want to put a photo here so no thanks...

Exercise (17). For each $f \in A$, let X_f denote the complement of $V(f)$ in $X = \mathrm{Spec}(A)$. The sets X_f are open. Show that they form a basis of open sets for the Zariski topology, and that

1. $X_f \cap X_g = X_{fg}$;
2. $X_f = \varnothing \iff f$ is nilpotent;
3. $X_f = X \iff f$ is a unit;
4. $X_f = X_g \iff \sqrt{(f)} = \sqrt{(g)}$;
5. X is quasi-compact (that is, every open covering of X has a finite subcovering).
6. More generally, each X_f is quasi-compact.
7. An open subset of X is quasi-compact if and only if it is a finite union of sets X_f.

The sets X_f are called *basic open sets* of $X = \mathrm{Spec}(A)$.

Solution: Note that $\mathfrak{p} \in X_f$ iff $f \notin \mathfrak{p}$. Every $\mathfrak{p} \in X$ is proper, and so some $f \in X - \mathfrak{p}$, so $\mathfrak{p} \in X_f$. Hence X_f covers X. Next, if $\mathfrak{p} \in X_f \cap X_g$, then $\mathfrak{p} \in X_{fg}$ as we will show next. Hence the X_f form a basis!

1. Note that since $\mathfrak{p}$ is prime, $f, g \notin \mathfrak{p}$ iff $fg \notin \mathfrak{p}$. Hence $X_f \cap X_g = X_{fg}$ directly!
2. $X_f = \varnothing$ iff $f \in \mathfrak{p}$ for every $\mathfrak{p} \in \operatorname{Spec}(A)$, iff f is in the nilradical.
3. If $X_f = X$ then f is not contained in any prime ideal. Hence it is not contained in any maximal ideal, so it is a unit. Conversely, suppose f is a unit. Then $(f) = A$, so $X_f = X$.
4. If $X_f = X_g$ then $f \in \mathfrak{p}$ iff $g \in \mathfrak{p}$. Recall that $\sqrt{(f)}$ is the intersection of all prime ideals containing (f). But a prime contains (f) iff it contains (g), so $\sqrt{(f)} = \sqrt{(g)}$. Conversely, if $\sqrt{(f)} = \sqrt{(g)}$, then $V(f) = V(\sqrt{(f)}) = V(\sqrt{(g)}) = V(g)$.
5. Let X be covered by $(X_{f_i})_{i \in I}$s. Then $\bigcap_{i \in I} V(f_i) = V(\sum_{i \in I}(f_i)) = \varnothing$ by Exercise (15). Since every maximal ideal is prime, this necessarily implies that $\sum_{i \in I}(f_i) = (1)$. Hence $1 = \sum_{j \in J} f_j g_j$ for some finite J. Then $\sum_{j \in J}(f_j) = 1$, so X is covered by $(X_{f_j})_{j \in J}$.
6. Let X_f be covered by $(X_{fg_i})_{i \in I}$. Then $V(\sum_{i \in I}(fg_i)) = V(f)$. Hence a prime ideal contains f iff it contains $\sum_{i \in I}(fg_i)$. Hence their radicals are equal. Hence $(f) \subseteq \sqrt{\sum_{i \in I}(fg_i)}$. This means that $f^n = \sum_{j \in J} fg_j h_j$ for some finite set j and some n. If $\mathfrak{p}$ is a prime ideal containing $\sum j \in J(fg_j)$, then $f^n \in \mathfrak{p}$, so $f \in \mathfrak{p}$. Hence $V(\sum_{j \in J}(fg_j)) \subseteq V(f)$. Thus $(X_{fg_j})_{j \in J}$ covers X_f.
7. Suppose $U \subseteq X$ is quasi-compact. Take $(X_{f_i})_{i \in I}$ so that $U = \bigcup_{i \in I} X_{f_i}$ from the basis. Then there is a finite subcover $(X_{f_j})_{j \in J}$. Hence U is a finite union of X_f sets.

   The converse is just a fact in topology, that the finite union of quasi-compact sets is quasi-compact. Indeed take a finite set $(X_{f_j})_{j \in J}$. Let $U = \bigcup_{j \in J} X_{f_j}$ and take a cover $(X_i)_{i \in I}$. Then the (X_i) cover each X_{f_j}, and hence each have a finite subcover. Their union is finite and covers U.

Exercise (18). For psychological reasons it is sometimes convenient to denote a prime ideal of A by a letter such as x or y when thinking of it as a point of $X = \operatorname{Spec}(A)$. When thinking of x as a prime ideal of A, we denote it by $\mathfrak{p}_x$ (logically, of course, it is the same thing). Show that

1. the set $\{x\}$ is closed (we say that x is a closed point) in $\operatorname{Spec}(A) \iff \mathfrak{p}_x$ is maximal;
2. $\overline{\{x\}} = V(\mathfrak{p}_x)$;
3. $y \in \overline{\{x\}} \iff \mathfrak{p}_x \subseteq \mathfrak{p}_y$;
4. X is a T_0-space (this means that if x, y are distinct points of X, then either there is a neighbourhood of x which does not contain y, or else there is a neighbourhood of y which does not contain x.

Solution:

1. If $\{x\}$ is closed then $\{x\} = V(\mathfrak{a})$ for some $\mathfrak{a}$. The set of prime ideals containing $\mathfrak{a}$ is just x, so necessarily $\mathfrak{a}$ is maximal and equal to $\mathfrak{p}_x$. Conversely if $\mathfrak{p}_x$ is maximal, then the set of ideals containing it is just $V(\mathfrak{p}_x) = \{\mathfrak{p}_x\} = \{x\}$.

2. A point y is in the closure of $\{x\}$ iff every open neighbourhood of y contains x. This is true iff it is true for every basis element. That is, for all $f \in A$, if $y \in X_f$ then $x \in X_f$. This is equivalent to that, if $f \notin \mathfrak{p}_y$ then $f \notin \mathfrak{p}_x$. Finally this is equivalent to saying $\mathfrak{p}_x \subseteq \mathfrak{p}_y$, i.e. $\mathfrak{p}_y \in V(\mathfrak{p}_x)$.

3. This is the definition of $V(\mathfrak{p}_x)$.

4. Notice that for distinct x, y, from part 3. that if $y \in \overline{\{x\}}$ then $x \notin \overline{\{y\}}$, and so we can take the open set $X - \overline{\{y\}}$ which contains x but not y. If $y \notin \overline{\{x\}}$, then $\mathfrak{p}_x \not\subseteq \mathfrak{p}_y$, hence $\mathfrak{p}_y \notin V(\mathfrak{p}_x)$. Hence we can take the open set $X - V(\mathfrak{p}_x)$ which contains y but not x.

Exercise (19). A topological space X is said to be *irreducible* if $X \neq \varnothing$ and if every pair of non-empty open sets of X intersect, or equivalently if every non-empty open set is dense in X. Show that $\mathrm{Spec}(A)$ is irreducible if and only if the nilradical of A is a prime ideal.

Solution: Note that X is irreducible iff every pair of non-empty open basis elements intersect. Equivalently, for all non-nilpotent $f, g \in A$, the product fg is not nilpotent by Exercise (17). This is exactly the statement of the nilradical being prime!

Exercise (20). Let X be a topological space.

1. If Y is an irreducible (Exercise (19)) subspace of X, then the closure $\overline{Y}$ of Y in X is irreducible.

2. Every irreducible subspace of X is contained in a maximal irreducible subspace.

3. The maximal irreducible subspaces of X are closed and cover X. They are called the *irreducible components* of X. What are the irreducible components of a Hausdorff space?

4. If A is a ring and $X = \mathrm{Spec}(A)$, then the irreducible components of X are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of A (Exercise (8)).

Solution:

1. Let U, V be non-empty open subsets of $\overline{Y}$. Then $U \cap Y, V \cap Y$ are non-empty. Hence they interesct, so U and V intersect.

2. If there are no irreducible subspaces of X then the proposition holds trivially. Else, we can apply Zorn's lemma. Let C be a chain of irreducible subspaces. Let U, V be open subsets of the union of C. Then there is some subspace Y containing both U and V. Hence, they intersect, so they intersect in the union. Therefore the union of C is irreducible, and thus is an upper bound of C.

3. If Y is a maximal irreducible subspace, then $\overline{Y}$ is irreducible, and hence equal to Y, so Y is closed. The maximal irreducible subspaces cover X since irreducible spaces cover X, indeed each point is irreducible and contained in some maximal irreducible subspace (this is given by Zorn's lemma).

   Since every pair of points in a Hausdorff space can be separated by disjoint open neighbourhoods, if a subspace has at least two points it is not irreducible. Hence the irreducible subspaces (and irreducible components) are the points, which indeed are closed!

4. Let Y be a closed irreducible subspace. Then it is closed, so it is equal to $V(\sqrt{\mathfrak{a}})$ for some ideal $\mathfrak{a}$. We first aim to show that $\sqrt{\mathfrak{a}}$ is prime. Suppose $fg \in \sqrt{\mathfrak{a}}$. Then $X_{fg} \cap Y = \emptyset$ so one of X_f or X_g is empty in Y. Assume X_f is. Then every prime ideal in Y contains f. Hence the intersection of said prime ideals contains f, which is $\sqrt{\mathfrak{a}}$. If instead X_g were empty, then g would be an element of $\sqrt{\mathfrak{a}}$. Hence $\sqrt{\mathfrak{a}}$ is prime. Conversely, consider $Y = V(\mathfrak{p})$ for some prime $\mathfrak{p}$ and let X_{fg} be empty in Y. Then $fg \in \mathfrak{p}$. Hence either $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$, so either X_f is empty in Y or X_g is, so Y is irreducible.

   Now suppose that $Y = V(\mathfrak{p})$ is an irreducible component. Then if $Y \subseteq Y' = V(\mathfrak{q})$ for some prime $\mathfrak{q}$, then $\mathfrak{q} \subseteq \mathfrak{p}$. However, $\mathfrak{p} = \mathfrak{q}$ since Y is maximal, so $\mathfrak{p}$ is minimal.

2 Modules

Exercise 2.2. 1. $\mathrm{Ann}(M+N) = \mathrm{Ann}(M) \cap \mathrm{Ann}(N)$

2. $(N : P) = \mathrm{Ann}((N+P)/N)$

Solution:

1. Note that $a(M+N) = aM + aN$. Hence $a \in \mathrm{Ann}(M+N)$ iff $aM + aN \subseteq 0$ iff $aM \subseteq 0$ and $aN \subseteq 0$ iff $a \in \mathrm{Ann}(M) \cap \mathrm{Ann}(N)$.

2. If $aP \subseteq N$ then every $p \in P$ has $ap \in N$. As $P \subseteq N + P$ this means $ap + N = 0 + N$, so $a \in \mathrm{Ann}((N+P)/N)$. Conversely, suppose $a(n+p) + N = 0 + N$ for all $n + p \in N + P$. Then $ap + N = 0 + N$, so $ap \in N$ and so $a \in (N : P)$.

Exercise 2.16. Let A, B be rings, let M be an A-module, P be a B-module and N an (A, B)-bimodule. Then $M \otimes_A N$ is naturally a B-module, $N \otimes_B P$ is naturally an A-module, and we have

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$$

Solution: I shall interpret the word naturally to mean that $- \otimes_A N$ is a functor from $\mathbf{Mod}_A$ to $\mathbf{Mod}_B$. It is sufficient to only show this direction of tensoring, the other direction should be identical. For each $b \in B$, define a bilinear map $(m, n) \mapsto m \otimes (bn)$. This induces a map from $M \otimes_A N \to M \otimes_A N$. This multiplication preserves addition and multiplication of the ring structure (indeed $m \otimes (b'(bn)) = m \otimes ((b'b)n)$ and $m \otimes ((b + b')n) = m \otimes (bn + b'n)$). Hence this makes $M \otimes_A N$ a B-module. Now suppose $f : M \to M'$ is an A-module homomorphism. We obtain a map $f \otimes N : M \otimes_A N \to M' \otimes N$ bilinearly mapping $(m, n) \mapsto f(m) \otimes n$. Indeed $\mathrm{id} \otimes N = \mathrm{id}$ and $(g \circ f) \otimes N = (g \otimes N) \circ (f \otimes N)$ (expand the definition).

I shall interpret this big isomorphism we have to prove as an (A, B)-bimodule isomorphism. These tensor products are special cases of a general bimodule tensor product described as follows. If M is an A-module and N is an (A, B)-bimodule, then $M \otimes_A N$ is an (A, B)-bimodule. It has the universal property that maps out of it are naturally isomorphic to A-bilinear maps that are B-linear in the right component. Indeed given such a bilinear map $f : M \times N \to P$, define $\hat{f} : M \otimes_A N \to P$ with A-bilinearity. A bimodule homomorphism is a map that is both an A and B-module homomorphism, and here $\hat{f}$ is already an A-module homomorphism. To see that it is a B-module homomorphism, observe

$$\hat{f}(b(m \otimes n)) = \hat{f}(m \otimes bn) = f(m, bn) = bf(m, n) = b\hat{f}(m \otimes n)$$

and extend by linearity. This construction is natural in P since post composition does not affect anything we have done. An identical construction can be made for an (A, B)-bimodule N and B-module P where we require A-linearity in the left component.

Now that we have a universal property, we can prove the big isomorphism using the yoneda lemma. To define a map out of $(M \otimes_A N) \otimes_B P$, we have to define a bilinear map out of $(M \otimes_A N) \times P$ that is A-linear in the left component. That is, we have to define a map out of $M \otimes_A N \times P$ that is a bimodule homomorphism with P fixed, and is B-linear with $M \otimes_A N$ fixed. This is the same thing as a map out of $M \times N \times P$ that is A-linear in M, B-linear in P and linear in both A and B in N. Reversing our steps, this shows that the universal property for $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are the same. Hence, they are isomorphic as (A, B)-bimodules.

Exercise 2.21. If $f : A \to B$ is a ring homomorphism and M is a flat A-module, then $M_B = B \otimes_A M$ is a flat B-module.

Solution: Let

$$0 \to N' \to N \to N'' \to 0$$

be an exact sequence of B-modules. Tensoring with M_B gives

$$0 \to (M \otimes_A B) \otimes_B N' \to (M \otimes_A B) \otimes_B N \to (M \otimes_A B) \otimes_B N'' \to 0$$

but this is isomorphic to the following sequence by exercise 2.16.

$$0 \to M \otimes_A (B \otimes_B N') \to M \otimes_A (B \otimes_B N) \to M \otimes_A (B \otimes_B N'') \to 0$$

This sequence is exact as a sequence of A-modules, but since kernels and images of maps are the same when regarded as a sequence of B-modules, this sequence is also exact as a B-module sequence. Hence M_B is flat.

Exercise (1). Show that $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0$ if m, n are coprime.

Solution: Let A be a Z-module (or abelian group) and let $f : (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z}) \to A$ be a bilinear map. Then $f(0,0) = f(m,0) = mf(1,0) = f(1,0)$ and $f(0,0) = f(0,n) = nf(0,1) = f(0,1)$. Hence, the induced map $\hat{f} : (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \to A$ is the zero map. Thus every map out of $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z})$ is the zero map, and $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \cong 0$.

Exercise (2). Let A be a ring, $\mathfrak{a}$ an ideal, M an A-module. Show that $A/\mathfrak{a} \otimes M$ is isomorphic to $M/\mathfrak{a}M$.

Solution: We will first show that $\mathfrak{a} \otimes M \cong \mathfrak{a}M$. Define $f : \mathfrak{a} \otimes M \to \mathfrak{a}M$ by $a \otimes m \mapsto am$. Similarly define $g : \mathfrak{a}M \to \mathfrak{a} \otimes M$ by $am \mapsto a \otimes m$. Both maps are homomorphisms, and are clearly self inverse.

Next consider the exact sequence

$$0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0.$$

Since tensor is right exact, we obtain the following exact sequence

$$\mathfrak{a}M \to M \to A/\mathfrak{a} \otimes M \to 0.$$

The left map is given from sending $am \mapsto am$ and so is injective. Hence we can extend the sequence to the left with a zero.

$$0 \to \mathfrak{a}M \to M \to A/\mathfrak{a} \otimes M \to 0$$

Thus $A/\mathfrak{a} \otimes M \cong M/\mathfrak{a}M$.